Project 13

Courier and Parcel Delivery Analytics System (CPDAS)

CSC 470: Software Engineering Lab
Department of Computer Science and Engineering
Project Specification   Group Size: 4–5 Students   Duration: One Semester

1 Project Overview

The Courier and Parcel Delivery Analytics System (CPDAS) is a desktop or web-based application used by the operations manager of a courier company in Dhaka. The booking counters already record every parcel with the customer's free-text delivery address, the riders' handheld app already records every scan (out for delivery, delivered, failed attempt) with time stamps, and the cash office already records cash-on-delivery (COD) collections and deposits. The purpose of this software is *not* to add, delete or modify records, but to **analyse** the stored data, perform computations, assign parcels to delivery zones by matching free-text addresses using an AI-based text-matching technique, and print operations reports.

The system reads four input data sets:

1. **Parcel data**: parcel ID, booking date and time, promised delivery hours (e.g. 48), the free-text delivery address, COD amount (zero for prepaid), assigned zone (blank until assigned).
2. **Scan data**: parcel ID, scan type (hub out, delivered, failed attempt), rider ID, date and time of the scan, failure reason text if failed.
3. **Rider data**: rider ID, name, home zone, duty days in the period.
4. **Zone catalogue data**: zone ID, hub, and a free-text description of the zone listing its areas, landmarks and common spelling variants (e.g. "Mirpur, Pallabi, Kazipara, Shewrapara, Section 1–14").

2 Module Structure

The system consists of five modules. Each module has a distinct purpose and performs computational operations only.

No.	Module	Purpose
M1	Delivery Performance Analytics	Compute delivery success rate, on-time percentage and average delivery time per zone.
M2	Rider Productivity Analytics	Compute parcels delivered per rider per day and each rider's failed-attempt rate.
M3	COD Reconciliation	Compute COD collected, deposited and any shortage per rider and per day.
M4	AI Address-to-Zone Matching	Match free-text delivery addresses to zone descriptions and assign the hub.
M5	Reporting	Generate and print all operations reports.

3 Module M1: Delivery Performance Analytics

3.1 Purpose

For any selected date and zone, this module computes how many parcels were delivered successfully, how many within the promised time, and the average time from hub-out to delivery.

3.2 Operations and Formulas

$$\text{Success}_z\,(\%) = \frac{n_z^{del}}{n_z^{att}} \times 100 \tag{1}$$

$$T_p = t_p^{del} - t_p^{book} \tag{2}$$

$$\text{OnTime}_z\,(\%) = \frac{|\{p : T_p \le P_p\}|}{n_z^{del}} \times 100 \tag{3}$$

$$\bar{T}_z = \frac{1}{n_z^{del}} \sum_{p \in z} T_p \tag{4}$$

where:

- Success_z = delivery success rate of zone z;
- n_z^{att} = parcels attempted in zone z on the selected date;
- n_z^{del} = parcels successfully delivered in zone z;
- T_p = total delivery time of parcel p in hours;
- t_p^{book} = booking time of parcel p; t_p^{del} = delivered-scan time of parcel p;
- P_p = promised delivery hours of parcel p (e.g. 48);
- $|\{p : T_p \le P_p\}|$ = the number of delivered parcels whose delivery time was within the promise;
- $\bar{T}_z$ = average delivery time of zone z.

3.3 Worked Example

Zone Z-MIR (Mirpur) on 08 July 2026: $n_z^{att} = 250$ parcels attempted, $n_z^{del} = 228$ delivered, of which 205 were within their promised hours; the delivery times of the 228 parcels summed to 7,524 hours:

$$\text{Success}_z = \frac{228}{250} \times 100 = 91.2\%, \qquad \text{OnTime}_z = \frac{205}{228} \times 100 = 89.9\%$$

$$\bar{T}_z = \frac{7524}{228} = 33.0 \text{ hours}$$

A zone whose success rate stays below 85% for three consecutive days is flagged for an *operations review* (extra rider or route change).

3.4 Input Form: Zone Performance Enquiry Form

Field	**Sample Input**
Zone	Z-MIR Mirpur (drop-down)
Date	08/07/2026
[Compute] [Print Report]	
Output: Success rate	91.2%
Output: On-time rate	89.9%
Output: Average delivery time	33.0 hours
Output: Operations review flag	No

4 Module M2: Rider Productivity Analytics

4.1 Purpose

This module measures each rider's output and reliability: how many parcels the rider delivers per duty day, and what share of the rider's attempts fail — so the manager can balance workloads and coach weak performers.

4.2 Operations and Formulas

$$\pi_r = \frac{n_r^{del}}{d_r} \tag{5}$$

$$\text{FailRate}_r\,(\%) = \frac{n_r^{fail}}{n_r^{del} + n_r^{fail}} \times 100 \tag{6}$$

where:

- π_r = productivity of rider r (parcels delivered per duty day);
- n_r^{del} = parcels the rider delivered in the period;
- d_r = duty days of the rider in the period;
- n_r^{fail} = failed delivery attempts recorded by the rider;
- FailRate_r = the rider's failed-attempt percentage.

The company average $\bar{\pi}$ is computed over all riders; a rider with $\pi_r < 0.7\,\bar{\pi}$ or $\text{FailRate}_r > 15\%$ is listed for coaching. The failure-reason texts ("customer unreachable", "address not found") are counted per reason so the manager can see *why* deliveries fail.

4.3 Worked Example

Rider RD-114 (Sumon Mia) in June 2026: $n_r^{del} = 508$ over $d_r = 24$ duty days, with $n_r^{fail} = 52$ failed attempts:

$$\pi_r = \frac{508}{24} = 21.2 \text{ parcels/day}, \qquad \text{FailRate}_r = \frac{52}{508 + 52} \times 100 = 9.3\%$$

The company average is $\bar{\pi} = 19.5$, so RD-114 is above average and his 9.3% fail rate is under the 15% line — no coaching flag. Of his failures, 31 were "customer unreachable" and 14 were "address not found".

5 Module M3: COD Reconciliation

5.1 Purpose

Riders collect cash on delivery and deposit it at the hub each evening. This module reconciles what each rider should have deposited against what was actually deposited, and computes any shortage — a daily financial control.

5.2 Operations and Formulas

$$C_r = \sum_{p \in \mathrm{del}(r)} m_p \tag{7}$$

$$\Delta_r = C_r - G_r \tag{8}$$

$$\mathrm{ShortageRate}_r\,(\%) = \frac{\sum_{days} \max(0, \Delta_r)}{\sum_{days} C_r} \times 100 \tag{9}$$

where:

- C_r = COD amount rider r collected on the day (the sum of the COD amounts m_p of the parcels the rider delivered);
- m_p = COD amount of parcel p (zero if prepaid);
- G_r = cash the rider actually deposited at the hub that evening;
- Δ_r = the day's difference: positive means a **shortage**, negative an over-deposit (usually a counting error);
- $\mathrm{ShortageRate}_r$ = the rider's shortage as a percentage of collections over the period.

5.3 Worked Example

On 08 July 2026, rider RD-114 delivered parcels with COD amounts totalling $C_r = BDT$ 38,450 and deposited $G_r = BDT$ 38,450: $\Delta_r = 0$, fully reconciled. Rider RD-131 collected BDT 27,300 but deposited BDT 26,300:

$$\Delta_r = 27300 - 26300 = BDT\ 1{,}000 \ \text{(shortage)}$$

Any $\Delta_r > 0$ is printed on the daily shortage report; a rider whose $\mathrm{ShortageRate}_r$ exceeds 0.5% over a month is escalated to the accounts manager, who takes the final decision.

6 Module M4: AI Address-to-Zone Matching

6.1 Purpose

Dhaka addresses are written irregularly — area names, landmarks and spelling variants mixed in free text. This AI-based module matches every parcel's address text against the description of each delivery zone and assigns the parcel to the right hub. An address whose similarity with a zone description is **at least 80%** is auto-assigned; otherwise the parcel goes to the dispatcher's manual-sorting queue. Every auto-assignment is recorded with its score, and the dispatcher can reassign any parcel.

6.2 Operation: Text Similarity by Cosine Measure

The address text and each zone description are converted into binary term vectors A and C after removing stop-words and generic address words (house, road, flat, floor):

$$\mathrm{sim}(A, C) = \frac{A \cdot C}{\|A\|\,\|C\|} \tag{10}$$

where:

- A = binary term vector of the parcel's delivery address;
- C = binary term vector of the zone description;

- $A \cdot C$ = dot product = number of terms common to both texts;
- $\|A\|$, $\|C\|$ = Euclidean norms, i.e. the square roots of the numbers of distinct terms in A and C respectively.

Groups may optionally upgrade the binary vectors to TF–IDF weights: $w_t = tf_t \times \log \frac{N}{df_t}$, where tf_t is the frequency of term t in the text, N is the total number of zone descriptions, and df_t is the number of descriptions containing term t. Because area names carry the signal while words like *house* and *road* appear in every address, the TF–IDF upgrade noticeably improves this module and is recommended for bonus credit.

6.3 Worked Example

Zone Z-MIR description terms: $C = \{mirpur, pallabi, kazipara, shewrapara, section\}$, so $\|C\| = \sqrt{5}$.

Parcel PC-77012's address reads: "*House 12, Road 3, Section 7, Pallabi, Mirpur, opposite Kazipara school.*" After removing stop-words and generic address words, the distinct terms are $A = \{section, pallabi, mirpur, kazipara, school\}$, so $\|A\| = \sqrt{5}$.

Common terms = $\{section, pallabi, mirpur, kazipara\}$, hence $A \cdot C = 4$.

$$\mathrm{sim}(A, C) = \frac{4}{\sqrt{5}\sqrt{5}} = \frac{4}{5} = 0.80 = 80\%$$

Since $80\% \geq 80\%$, parcel PC-77012 is auto-assigned to **Zone Z-MIR (Mirpur hub)**. The same address scores only 16% against the Uttara zone description, confirming the assignment. An address reading only "near college gate, Dhaka" scores below the threshold against every zone and is placed in the dispatcher's manual-sorting queue.

6.4 Input Form: Address Matching Form

Field	Sample Input
Parcel ID	PC-77012
Address text (read-only)	House 12, Road 3, Section 7, Pallabi, Mirpur…
	Run AI Matching
Output: Assigned zone	Z-MIR — Mirpur hub (80%)
Output: Second best	Z-UTT — Uttara (16%)
Output: Dispatcher override	Available

7 Module M5: Reporting

All reports are printable on A4 with the company header, report title, date and page number. Sample formats with values follow.

7.1 Report R1: Daily Delivery Report

Dhrutogoti Courier — Daily Delivery Report
Date: 08 July 2026

Zone	Attempted	Delivered	Failed	Success	On Time
Z-MIR Mirpur	250	228	22	91.2%	89.9%
Z-UTT Uttara	198	186	12	93.9%	92.5%
Z-DHN Dhanmondi	174	148	26	85.1%	84.5%
Totals	622	562	60	90.4%	89.5%

Top failure reason today: customer unreachable (34 of 60)

7.2 Report R2: Zone Performance Report

Dhrutogoti Courier — Zone Performance Report, Week 27, 2026				
Zone	**Avg. Delivery Time ($\bar{T}_z$)**	**Success (7-day)**	**Volume**	**Flag**
Z-UTT Uttara	29.4 h	94.1%	1,310	–
Z-MIR Mirpur	33.0 h	90.8%	1,702	–
Z-DHN Dhanmondi	41.6 h	83.9%	1,166	**Operations review (day 3)**
Recommendation printed: one additional rider for Dhanmondi.				

7.3 Report R3: Rider Productivity Report

Dhrutogoti Courier — Rider Productivity Report, June 2026						
Rider	**Name**	**Delivered**	**Duty Days**	**π_r/day**	**FailRate**	**Note**
RD-114	Sumon Mia	508	24	21.2	9.3%	Above average
RD-108	Alamgir Hossain	472	25	18.9	11.4%	–
RD-131	Liton Sarker	296	23	12.9	17.8%	**Coaching list**
Company average $\bar{\pi}$: 19.5/day Coaching threshold: $\pi_r < 13.7$ or FailRate > 15%						

7.4 Report R4: Daily COD Reconciliation Report

Dhrutogoti Courier — Daily COD Reconciliation, 08 July 2026 (BDT)					
Rider	**Name**	**Collected (C_r)**	**Deposited (G_r)**	**Δ_r**	**Status**
RD-114	Sumon Mia	38,450	38,450	0	Reconciled
RD-108	Alamgir Hossain	31,220	31,220	0	Reconciled
RD-131	Liton Sarker	27,300	26,300	**1,000**	**Shortage — accounts notified**
COD collected today: 96,970 Deposited: 95,970 Shortage: 1,000 (1 rider)					

7.5 Report R5: AI Zone Assignment Log

Dhrutogoti Courier — AI Zone Assignment Log, 08 July 2026				
Parcel ID	**Address (first words)**	**Assigned Zone**	**Similarity**	**Status**
PC-77012	House 12, Road 3, Section 7, Pallabi...	Z-MIR Mirpur	80%	Auto-assigned
PC-77013	Sector 11, Uttara, near Rajlokkhi...	Z-UTT Uttara	91%	Auto-assigned
PC-77020	near college gate, Dhaka	–	42%	Manual sorting queue
Parcels booked today: 655 Auto-assigned: 601 (91.8%) Manual queue: 54 Dispatcher overrides: 3				

8 Deliverables and Assessment

1. Software Requirements Specification (use cases, functional and non-functional requirements) — 15%
2. Design documents: context diagram, DFD (level 0 and 1), ER diagram, class diagram — 20%
3. Working implementation of modules M1–M5 with the formulas above — 35%
4. AI address-matching module with a demonstration on at least fifteen sample addresses across three zones (TF–IDF upgrade earns bonus credit) — 15%
5. Printed reports R1–R5 with real computed values, and final viva — 15%

Constraint reminder: the system must not implement add, delete or update operations; all marks are awarded for analytical computation, AI-based matching and reporting.